Quiz2

Q1. $X_1 \sim N(0,1)$, $X_t | X_{t-1} \sim N(\rho X_{t-1}, 1-\rho^2)$, $t = 2, \dots, T$.

$\hookrightarrow$ Markov.

$$\pi(\underset{\sim}{x}) = \pi(x_1, x_2, \dots, x_T)$$

$$= \pi(x_1)\pi(x_2|x_1)\pi(x_3|x_2)\cdots\pi(x_T|x_{T-1})$$

$$= \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x_1^2} \times \prod_{t=2}^{T} \frac{1}{\sqrt{2\pi(1-\rho^2)}} e^{-\frac{1}{2(1-\rho^2)}(x_t - \rho x_{t-1})^2}$$

$$= \frac{1}{(2\pi)^{T/2}(1-\rho^2)^{\frac{T-1}{2}}} e^{-\frac{1}{2}\left\{x_1^2 + \frac{1}{(1-\rho^2)}\sum_{t=2}^{T}(x_t - \rho x_{t-1})^2\right\}}.$$

$$\pi(x_t | x_1, \dots, x_{t-1}, x_{t+1}, \dots, x_T)$$

$$= \frac{\pi(x_1, x_2, \dots, x_{t-1}, x_t, x_{t+1}, \dots, x_T)}{\pi(x_1, \dots, x_{t-1}, x_{t+1}, \dots, x_T)}$$

$$\propto \pi(x_1, x_2, \dots, x_{t-1}, x_t, x_{t+1}, \dots, x_T) = \pi(x_1)\prod_{t=2}^{T}\pi(x_t|x_{t-1})$$

$$\propto \pi(x_t|x_{t-1})\,\pi(x_{t+1}|x_t).$$ [other terms do not involve x_t]

$$\propto e^{-\frac{1}{2(1-\rho^2)}(x_t - \rho x_{t-1})^2} \times e^{-\frac{1}{2(1-\rho^2)}(x_{t+1} - \rho x_t)^2}$$

$$\propto e^{-\frac{1}{2(1-\rho^2)}\left\{(x_t - \rho x_{t-1})^2 + (x_{t+1} - \rho x_t)^2\right\}}$$

$$\propto e^{-\frac{1}{2(1-\rho^2)}\left\{x_t^2 - 2\rho x_t x_{t-1} + \rho^2 x_{t-1}^2 + x_{t+1}^2 - 2\rho x_t x_{t+1} + \rho^2 x_t^2\right\}}$$

$$\propto e^{-\frac{1}{2(1-\rho^2)}\left\{(1+\rho^2)x_t^2 - 2x_t(\rho x_{t-1} + \rho x_{t+1})\right\}}$$

$$\propto e^{-\frac{1}{2\cdot\frac{1-\rho^2}{1+\rho^2}}\left\{x_t^2 - 2x_t \cdot \frac{\rho(x_{t-1} + x_{t+1})}{1+\rho^2}\right\}}$$

$$\Rightarrow x_t | x_1, \dots, x_{t-1}, x_{t+1}, \dots, x_T \sim N\left(\frac{\rho}{1+\rho^2}(x_{t-1} + x_{t+1}), \frac{1-\rho^2}{1+\rho^2}\right)$$

Q2. $\pi(\lambda \mid y_1,\dots,y_T) \propto \pi(y_1,\dots,y_T \mid \lambda)\,\pi(\lambda)$.

$y_1,\dots,y_T \mid \lambda \overset{\text{IID}}{\sim} \text{Poi}(\lambda)$.

$$\Rightarrow \pi(y_1,\dots,y_T \mid \lambda) = \prod_{t=1}^{T} \pi(y_t \mid \lambda) = \prod_{t=1}^{T} e^{-\lambda}\cdot \frac{\lambda^{y_t}}{y_t!}$$

$$= e^{-\lambda T}\cdot \lambda^{\sum_{t=1}^{T} y_t}$$

$\lambda \sim \text{Ga}(a,b) \Rightarrow \pi(\lambda) \propto \lambda^{a-1} e^{-b\lambda}$

$$\Rightarrow \pi(\lambda \mid y_1,\dots,y_T) \propto e^{-\lambda T}\cdot \lambda^{\sum_{t=1}^{T} y_t} \times \lambda^{a-1} e^{-b\lambda}$$

$$\propto e^{-\lambda(T+b)}\cdot \lambda^{a+\sum_{t=1}^{T} y_t - 1}.$$

$$\Rightarrow \lambda \mid y_1,\dots,y_T \sim \text{Gamma}\left(a+\sum_{t=1}^{T} y_t,\ b+T\right).$$

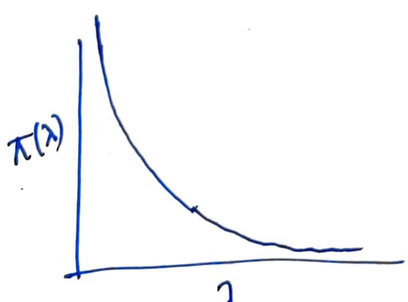
Q3. $Y \mid \lambda \sim \text{Poi}(\lambda)$

$L(\lambda) = e^{-\lambda}\cdot \lambda^{Y}/Y! \Rightarrow \ell(\lambda) = -\lambda + Y\log(\lambda) - \log(Y!)$.

$$\frac{\partial}{\partial\lambda}\ell(\lambda) = -1 + \frac{Y}{\lambda} \Rightarrow \frac{\partial^2 \ell(\lambda)}{\partial\lambda^2} = -\frac{Y}{\lambda^2} < 0.$$

$$I(\lambda) = E\left(-\frac{\partial^2 \ell(\lambda)}{\partial\lambda^2}\right) = E\left(+\frac{Y}{\lambda^2}\right) = \frac{1}{\lambda^2}E(Y) = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}.$$

$\pi(\lambda) \propto \sqrt{I(\lambda)}. \Rightarrow \pi(\lambda) \propto \lambda^{-1/2},\ \lambda > 0.$ $\boxtimes$



c = prop. Constant

$$\int_0^\infty \pi(\lambda)\,d\lambda = \int_0^\infty c\lambda^{-1/2}\,d\lambda$$

$$= c\int_0^\infty \lambda^{-1/2}\,d\lambda = c.\ \frac{\lambda^{-1/2+1}}{-1/2+1}\Bigg|_0^\infty$$

$= \infty$. $\Rightarrow$ Improper prior.

$$\pi(\lambda \mid Y) \propto \pi(Y \mid \lambda)\,\pi(\lambda) \propto e^{-\lambda}\cdot\lambda^{Y}\cdot\lambda^{-1/2} \propto \lambda^{-1/2+Y} e^{-\lambda} \propto e^{-\lambda}\cdot\lambda^{Y+\frac{1}{2}-1}$$

$\Rightarrow \lambda \mid Y \sim \text{Gamma}(Y+1/2,\ 1)$, So proper $\forall\ Y = 0,1,2,\dots$

Q4. $f(Y|\theta) \geq c(Y)$. $\forall \theta$.

prior is improper $\Rightarrow \int \pi(\theta) d\theta = \infty$.

$\Rightarrow \pi(\theta|y) \propto f(Y|\theta).\pi(\theta)$.

C_0 = prop. constant

$\Rightarrow \pi(\theta|Y) = C_0 f(Y|\theta)\pi(\theta)$

$$\Rightarrow \int \pi(\theta|Y) d\theta = \int C_0 f(Y|\theta)\pi(\theta) d\theta.$$

$$= C_0 \int f(Y|\theta)\pi(\theta) d\theta.$$

$$\geq C_0 \int c(Y)\pi(\theta) d\theta.$$

$$= C_0 . c(Y) . \int \pi(\theta) d\theta.$$

$$= \infty.$$

$\Rightarrow \int \pi(\theta|Y) d\theta = \infty \Rightarrow$ posterior is improper.

Q5. Likelihood : $f(Y|\theta) = L(\theta|Y)$.

prior : $\pi(\theta) = \sum_{l=1}^{L} w_l \pi_l(\theta)$, where w_l are mixture weights and $\pi_l(\theta)$ are component-wise priors.

Now, π_l are conjugate with $f(Y|\theta)$.

$$\text{Now, } \pi(\theta|Y) = \frac{f(Y|\theta)\pi(\theta)}{\int f(Y|\theta)\pi(\theta) d\theta} = \frac{f(Y|\theta)\pi(\theta)}{f(Y)}$$

$$= \frac{f(Y|\theta).\sum_{l=1}^{L} w_l \pi_l(\theta)}{f(Y)} = \sum_{l=1}^{L} w_l . \frac{f(Y|\theta).\pi_l(\theta)}{f(Y)}.$$

$$= \sum_{l=1}^{L} w_l . \frac{1}{f(Y)} . \int f(Y|\theta)\pi_l(\theta) d\theta . \frac{f(Y|\theta)\pi_l(\theta)}{\int f(Y|\theta)\pi_l(\theta) d\theta}.$$

$$= \sum_{l=1}^{L} w_l' \cdot \frac{f(y|\theta)\pi_l(\theta)}{\int f(y|\theta)\pi_l(\theta)d\theta} = \sum_{l=1}^{L} w_l' \pi_l(\theta|y).$$

where $w_l' = w_l \cdot \dfrac{\int f(y|\theta)\pi_l(\theta)d\theta}{\int f(y|\theta)\pi(\theta)d\theta}$.

Now, $\pi_l(\theta)$ is Conjugate with $f(y|\theta)$. Hence, $\pi_l(\theta|y)$ belongs to the same family ~~with~~ of $\pi_l(\theta)$.

thus, the posterior is also a mixture of L Components where each Component belongs to the same family as the prior.

$\Rightarrow \pi(\theta) = \sum_{l=1}^{L} w_l \pi_l(\theta)$ is Conjugate in this Case.